



Timing of repair of congenital diaphragmatic hernia in patients supported by extracorporeal membrane oxygenation (ECMO)



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ABSTRACT

Purpose: The optimal timing of repair for congenital diaphragmatic hernia (CDH) in patients requiring extracorporeal membrane oxygenation (ECMO) is controversial. Repair during ECMO may improve respiratory function by restoring normal anatomy. However, there is increased risk of complications including surgical bleeding. The purpose of this study was to examine the impact of timing of CDH repair on outcomes in a large cohort of patients treated at a single institution.

Methods: We retrospectively reviewed charts of all CDH patients in our Pulmonary Hypoplasia Program from 2004 to 2013. Categorical variables were analyzed by Fisher's exact test and continuous variables by Mann–Whitney t-test ($p < 0.05$).

Results: A total of 77 CDH patients required ECMO support during the study dates. Of these, 16 patients did not survive to repair, 3 patients were repaired prior to cannulation, 41 patients were repaired during ECMO, and 17 patients were repaired after decannulation. Survival was 67%, 43.9%, and 100% for those repaired prior to, during, or post ECMO, respectively, with statistical significance associated with repair after decannulation ($P < 0.0001$). Operative bleeding requiring transfusion occurred in 12 patients repaired on ECMO, while no significant bleeding occurred in patients repaired after decannulation ($P = 0.003$).

Conclusion: Outcomes were improved in CDH patients undergoing surgical repair following ECMO with significantly increased survival, lower rates of surgical bleeding, and decreased total duration of ECMO therapy compared to patients repaired on ECMO. In patients who can be successfully weaned from ECMO, our study supports a role for delayed repair off ECMO with reduced operative morbidity and increased survival.

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Congenital diaphragmatic hernia (CDH) is a severe congenital anomaly affecting 1 in every 2500–3000 live births. Although advances in surgical technique and neonatal care have led to improved overall survival rates, pulmonary hypoplasia and persistent pulmonary hypertension (PH) are frequently observed in CDH patients and remain the most significant factors determining survival [1]. In cases of severe PH refractory to vasodilatory therapy, ECMO may be required to achieve adequate gas exchange and perfusion. Rates of ECMO usage in CDH vary from 0% to 61% in the literature [2,3], with mortality rates averaging 30%–40% in these patients [2,4].

ECMO cannulation is frequently required prior to CDH repair, raising the question of optimal timing of repair in such patients. Repair during ECMO may improve respiratory function by restoring normal anatomy and relieving intrathoracic compression, however the requirement for systemic anticoagulation during ECMO support portends an increased risk of operative complications including surgical bleeding. The purpose

of this study was to examine the impact of timing of CDH repair on outcomes in a large cohort of patients treated at a single institution.

1. Materials and methods

The records of all CDH survivors treated at the Children's Hospital of Philadelphia and enrolled in the Pulmonary Hypoplasia Program between January 2004 and December 2012 were retrospectively analyzed under institutional IRB approval (IRB 06-003779). Recorded data on patient demographics, prenatal course and imaging, surgical resection and postoperative course were analyzed. Postoperative surgical site bleeding was defined as the requirement for peri- or postoperative blood transfusion. Prenatal CDH severity was assessed by liver position and lung to head ratio (LHR) on ultrafast fetal MRI and detailed sonographic evaluation as previously described [5]. Indication for initiation of ECMO therapy included failure of medical management to avoid ventilator related lung injury or persistent hypotension/acidosis. Inclusion criteria included cannulation for ECMO as well as surgical repair of CDH, while exclusion criteria included death prior to CDH repair.

Data are described as median (range). Statistical analysis was performed using Fisher's exact test for categorical variables and Mann–Whitney test for continuous variables. $P \leq 0.05$ was considered

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significant. All data analysis was conducted using GraphPad Prism 6.0 (La Jolla, CA).

2. Results

From January 2004 through December 2012, a total of 77 CDH patients enrolled in the Pulmonary Hypoplasia Program were placed on ECMO support. 16 patients did not survive to repair and were excluded from further study, 3 patients were repaired prior to cannulation, 41 patients were repaired during ECMO, and 17 patients were repaired after decannulation. Of the 16 non-survivors, care was withdrawn in 10 following a diagnosis of catastrophic intracerebral hemorrhage, while in 6, care was withdrawn due to multiorgan failure refractory to medical management.

Table 1 summarizes the baseline characteristics of the study patients stratified according to timing of CDH repair. Patients repaired during ECMO therapy were compared to patients repaired either prior to cannulation or following decannulation. There were no statistically significant differences in gestational age, birth weight, side of CDH defect, or liver herniation in patients repaired either prior to or following ECMO therapy compared to patients repaired on ECMO. Patients repaired prior to ECMO cannulation did have statistically significantly higher average LHR levels compared to patients repaired on ECMO, however there was no difference in the average LHR values between patients repaired following ECMO treatment compared to those repaired on ECMO.

Table 2 summarizes the clinical characteristics of the study patients stratified according to timing of repair. As expected, patients requiring ECMO cannulation following CDH repair were repaired significantly earlier (median day of repair 4 (3–5) days versus 12 (1–30) days, $P < 0.01$) and were placed on ECMO significantly later (median age at cannulation 5 (3–8) days versus 1 (1–7) day, $P < 0.01$) than patients repaired on ECMO. Similarly, patients repaired following decannulation were repaired significantly later (median day of repair 25 (9–35) versus 12 (1–30) days $P < 0.01$), but were cannulated at similar ages (median age at cannulation 1 (1–7) days versus 1 (1–7) days, $P = 0.24$). The requirement for patch repair was also significantly lower in patients repaired prior to cannulation ($P = 0.04$), but did not differ significantly in patients repaired on or following ECMO ($P = 0.22$). The duration of ECMO therapy was significantly shorter in patients repaired after decannulation compared to those repaired on ECMO ($P = 0.01$) but length of stay was not significantly different between any of the cohorts studied.

No patients repaired prior to cannulation or following decannulation experienced significant surgical site bleeding, as defined by the requirement for peri- or post-operative transfusion. Twenty-nine percent (12/41) of patients repaired on ECMO did require transfusion for significant surgical bleeding. Patients with significant surgical bleeding had numerous complications, including the requirement for temporary abdominal closure in seven patients, disseminated intravascular coagulopathy (DIC) in one patient, and requirement for thoracotomy in one patient. Overall survival was 66%, 44%, and 100% for patients repaired prior to, during, or post ECMO respectively, with statistical significance associated with repair after decannulation ($P < 0.01$).

3. Discussion

Although survival rates for CDH have improved significantly with advances in neonatal care, long-term survivors face substantial risk of morbidity and mortality which is largely attributable to pulmonary hypoplasia and PH. In this retrospective series of 77 patients with CDH requiring ECMO therapy for severe PH, we observed improved outcomes in patients repaired following successful weaning from ECMO, with significantly increased survival, lower rates of surgical bleeding, and decreased duration of ECMO runs compared to patients repaired on ECMO. In patients who can be successfully weaned from ECMO, our study supports a role for delayed repair off ECMO with reduced operative morbidity and increased survival.

Much controversy exists in the literature regarding the optimal timing of CDH repair in patients requiring ECMO. In the general CDH population, repair was traditionally treated as a surgical emergency, with prompt surgical intervention felt to confer advantages including improved venous return, restoration of normal anatomy, and reduced mass effect on the thoracic cavity [6]. With advances in neonatal intensive care including protective ventilation and permissive hypercapnia strategies to minimize iatrogenic injury, delayed repair may offer advantages with respect to patient stabilization prior to repair. Furthermore, repair during ECMO has been reported to be associated with significant bleeding risk due to the requirement for systemic anticoagulation [7].

Dassinger et al. explored the question of timing of CDH repair in 48 patients supported on ECMO [8]. In their study, survival of patients repaired within 72 h of commencement of ECMO therapy resulted in survival rates of 71% with perioperative bleeding requiring intervention in 8.8% of patients. Fallon et al. reviewed outcomes following early (<72 h), late (>72 h) or post-decannulation repair, and observed similar survival rates between all three cohorts with significantly shorter ECMO runs in patients in the early repair cohort [9]. Our findings differ in both the demonstrated survival advantage in post-decannulation repairs as well as the shorter duration of ECMO runs in this patient cohort. While these findings could be attributed to lesser disease severity in the post-decannulation cohort in our study, the lack of significant differences in prenatal markers of disease severity such as liver position and LHR, as well the equal rates of requirement for patch repair across all cohorts, argues against a more favorable disease phenotype in the post-decannulation repair cohort. The ability to successfully wean from ECMO may independently reflect improved pulmonary function in such patients which has not occurred in those unable to be weaned from ECMO before repair. The increased length of ECMO runs and decreased survival of patients repaired on ECMO thus likely reflect a sicker patient population in whom long ECMO runs were undertaken awaiting improvement in pulmonary function which did not occur. Once a bleeding complication occurs, there is often a vicious cycle of fluid overload, hemodynamic swings, and increased risk of secondary complications.

Interestingly, increased bleeding complications were not observed in Fallon et al.'s more recent study, and the authors suggest the use of an expanded coagulation profile for more accurate management of anticoagulation titration during repair as a potential explanation [9]. While activated clotting time (ACT) measurements alone have traditionally been employed to monitor anticoagulation therapy during ECMO,

Table 1
Baseline characteristics of CDH patients.

	Gestational Age Median (range)	Birthweight Median (range)	Side of defect: Left (%)	LHR Median (range)	Liver position: intrathoracic (%)
Repaired pre-ECMO (n = 3)	37 (34–40)	3000 (2500–3430)	2 (66%)	1.7 (1.4–2.0)	1 (33%)
P value	0.77	0.87	0.55	0.02	0.18
Repaired on ECMO (n = 41)	36 (29–40)	3000 (2280–4000)	32 (78%)	1.05 (0.58–1.6)	31 (76%)
Repaired post-ECMO (n = 17)	36 (30–40)	3150 (2600–4649)	13 (77%)	1.0 (0.63–1.9)	11 (65%)
P value	1.00	0.18	1.00	0.30	0.74

P values compare patients "Repaired pre-ECMO" and "Repaired post-ECMO" to patients "Repaired on ECMO".

Table 2

Details of ECMO runs.

	Age at repair (days) Median (range)	Age @ cannulation (days) Median (range)	Patch repair (%)	Surgical Site Bleeding (%)	Length of ECMO (hours) Median (range)	LOS (days) Median (range)	Survival to discharge (%)
Repaired pre-ECMO (n = 3)	4 (3–5)	5 (3–8)	1 (33%)	0 (0%)	309 (250–600)	83 (78–88)	2 (66%)
P value	<0.01	<0.01	0.04	0.55	0.54	0.71	0.58
Repaired on ECMO	12 (1–30)	1 (1–7)	37 (90%)	12 (29%)	452 (126–703)	68 (17–223)	18 (44%)
Repaired post-ECMO	25 (9–35)	1 (1–7)	13 (76%)	0	311 (166–434)	98 (48–319)	17 (100%)
P value	<0.01	0.24	0.22	<0.01	0.02	0.19	<0.01

P values compare patients “Repaired pre-ECMO” and “Repaired post-ECMO” to patients “Repaired on ECMO”.

increasing evidence supports a role for monitoring additional factors such as prothrombin time, fibrinogen, heparin levels, and anti-thrombin III levels [10]. All patients included in this study were monitored with serial ACT measurements. Since late 2011, we have also followed antiXa, prothrombin time, partial thromboplastin, and antithrombin 3 in addition to ACT measurements as part of ELSO anticoagulation task force. Thromboelastography is also used in selected cases.

The results of this single-center retrospective study demonstrate an overall survival rate of 60% in CDH patients requiring ECMO therapy prior to, during or following surgical repair, and suggest a reduction in operative complications and improved survival in patients repaired after ECMO support. These findings support a role for delayed repair off ECMO when patient characteristics allow them to be weaned from support, although continued prospective study is necessary to further elucidate the factors determining improved outcomes in these patients.

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